DISCRETE TRANSISTOR OP-AMP

Jason Gallicchio and Claude, rev A, 2026-10-24

https://github.com/gallicchio/op-amp-pcb
circuit, scripts, schematic, this board

The op-amp of Learning the Art of Electronics
(Hayes & Horowitz, 2016), built from eight
2N3904 / 2N3906 so you can probe every node.

POWER: +15 V, GND, −15 V into the terminal (±16 V max), 20 mA.
  Set the supply's current limit to 50 mA.
  Both LEDs light if it is right.
SWITCHES, usual positions:
  TAIL 1 on: the Q6 current source.
  COMPENSATION 1 on: 100 pF + 1k.
  FEEDBACK as needed (below).
  TAIL 2 instead: a 10k tail resistor.
  COMPENSATION 2 instead: the bare 100 pF, no 1k (the book's):
    the x11 amp still works, a follower rings.
FEEDBACK:
  1 = X to IN−: gain 11.
  2 = OUT to IN−: follower.
  BOTH OFF: a bare op-amp.
AS AN OP-AMP IN YOUR OWN CIRCUIT (Labs 6-9):
  FEEDBACK both off.
  IN−, IN+, OUT and GND on the terminals are pins 2, 3, 6 of a '411.
  Mind its 2 uA of input bias current
    (10 M feedback for photodiode will not work; everything else should).
  Its output limits at 17 mA, like a real one (Q7, Q8: not in the book).
OFFSET NULL: as a follower with IN+ grounded, turn RV1 until OUT reads 0 V.
INPUTS: ±12 V.
OUT: ±12 V into 1k, 15 mA, through R17's 47 ohm (outside the loop).
  A short to ground is survived; a short to a rail, only for seconds.

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